

Ali Hassan

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About Me

Master's student in Embedded Systems Security at the University of Grenoble with expertise in hardware design, firmware development, and system integration. Skilled in C/C++, Python, and MATLAB with experience in microcontroller programming (STM32, ESP32, nRF, PIC) and real-time operating systems like FreeRTOS and Zephyr. Experienced in IoT development using BLE, Zigbee, LoRaWAN, Wi-Fi, and CAN, and in FPGA design, RTL implementation, and multilayer PCB design focusing on signal integrity and EMI optimization. Passionate about embedded security, hardware-software co-design, and seeking an internship to apply and expand expertise in embedded systems and IoT security.

Education

Master's in Embedded Systems Security

Grenoble INP Esisar, Valence, France

Sep 2025 – Present

BS Electrical Engineering

COMSATS University Islamabad, Pakistan

Sep 2020 – Aug 2024

Technical Skills

- **Embedded Systems:** Firmware Development, FreeRTOS, IoT Integration, Hardware Design, Design Verification & Validation
- **Hardware Platforms & Controllers:** STM32, nRF, ESP32, PIC, Raspberry Pi, Zybo Board, FPGA, PLC
- **Communication Protocols:** UART, SPI, I²C, CAN, USB, BLE, Zigbee, LoRaWAN, Wi-Fi
- **Embedded Security:** Secure Boot, Cryptography, Hardware Security, Trusted Execution Environments (TEE)
- **Programming Languages:** C, C++, Python, MATLAB, VHDL
- **Software Tools:** Altium Designer, Keil uVision, STM32CubeIDE, MPLAB X, Arduino IDE, MATLAB/Simulink, Git/GitHub
- **Soft Skills:** Problem Solving, Critical Thinking, Adaptability, Communication, Time Management

Experience

Embedded Systems Engineer, MindTune Innioations

- Developed real-time IoT embedded systems using Zephyr RTOS and FreeRTOS.
- Built a wireless EEG system acquiring data from ADS1299 and transmitted over BLE using nRF52840.
- Designed wireless earbuds on Jieli AC6966 SoC, integrating modern features and optimizing firmware for low-latency, power-efficient performance.

Wah Cantt, PK
Mar 2025 – Sep 2025

Embedded Systems Engineer, Revive Medical Technologies

- Developed firmware for PIC, STM32, and nRF microcontrollers for reliable medical devices.
- Implemented UART, SPI, I²C, and IoT-based communication for seamless device interoperability.
- Integrated sensors and communication modules for remote monitoring.

Islamabad, PK
Aug 2024 – Mar 2025

Projects

Catheter Trackability Testing Machine

- Developed a medical testing machine integrating a Futek load cell and stepper motor for precise force measurement.
- Tools: Microcontroller, Embedded C, Futek Load Cell, Motor Control, Sensor Integration.

[GitHub](#)

Disposable Real-Time Pressure Monitoring Device

- Engineered a medical device for real-time physiological pressure measurement and monitoring.
- Tools: STM32, Arduino, PCB Design, Real-Time Acquisition, Sensor Calibration.

[GitHub](#)

Robotic Arm-Based Skin Tumor Detection and Intervention System using Computer Vision

- Designed a robotic arm system for automated skin tumor detection and intervention using vision algorithms.
- Tools: Raspberry Pi, Python, Computer Vision, Image Processing, Machine Learning.

[GitHub](#)

ISE to Modbus RTU Converter

- Designed a 4-layer PCB with isolated power, 24-bit ADC, and RS-485 interface for industrial communication.
- Tools: PCB Design, Schematic Capture, Signal Integrity, Isolation Design, Embedded Hardware.

[GitHub](#)

USB Audio Dongle

- Implemented a USB Audio Class 1.0 device on NXP MCXC444 for low-latency full-duplex audio streaming.
- Tools: Embedded C, USB Protocol, Audio Processing, Descriptor Design, MCU Programming.

[GitHub](#)

Languages

English: (Fluent)

French: (Basic)